



DPUHeli

Vehicle dynamics software, which is not modeled after a real vehicle but well suited for testing of scenarios and for control with a joystick.

1. Inputs:

<i>Pov</i>	Specifies the gaze direction. (e.g. controlled by a coolie hat on a joystick.)
<i>Slider</i>	Specifies the altitude above the ground, in which the camera is. (e.g. controlled by Throttle on a joystick.)
<i>rotz</i>	Additional yaw of the camera. (e.g. controlled by Z-rotation on a joystick.)
<i>Gear</i>	Gear shift. -1 = backwards, 0 = neutral, 1 = forwards
<i>SliderReset</i>	If $\neq 0$, camera altitude (slider) is set back to the default val-ue.
<i>YawBoost</i>	If $\neq 0$, through <i>rotz</i> even the vehicle is rotated (output yaw). If $= 0$, through <i>rotz</i> only the camera is rotated (output ViewYaw).
<i>SteeringWheel</i>	Steering angle. [rad]
<i>AcceleratorPedal</i>	Accelerator pedal position from 0 ... 1
<i>BrakePedal</i>	Brake pedal position from 0 ... 10
<i>XOffset, YOffset, ZOffset</i>	Additional translation of the vehicle in the simulation's coordinate system during the simulation [m]. This input is computed by DPUSCNX. Default: 0.0 m
<i>PitchRoad</i>	Tilt of road in x direction [rad] (output of DPUSCNX)
<i>RollRoad</i>	Tilt of road in y direction [rad] (output of DPUSCNX)
<i>ZRoad</i>	Height of road surface [m] (output of DPUSCNX)

2. Parameters:

<i>l_VR</i>	Distance from center of gravity to front wheel (along X axis) [m]
<i>l_HR</i>	Distance from center of gravity to rear wheel (along X ax-is) [m]
<i>c_VR</i>	Inertia of front axis
<i>c_HR</i>	Inertia of rear axis
<i>M_Fzg</i>	Vehicle weight [kg]
<i>i_SW</i>	Inertia during steering

3. Static parameters:

<i>SetupX, SetupY, SetupZ</i>	Position of the setup point in the global coordinate system [m]. This input is computed by DPUSCNX.
<i>SetupYaw, SetupPitch, SetupRoll</i>	Rotation of the setup point in the global coordinate sys-tem [rad]. This input is computed by DPUSCNX.
<i>SetupDone</i>	Flag specifying if DPUSCNX has completely setup the vehicle.



4. Outputs:

<i>v</i>	Speed. [m/s]
<i>v_kmh</i>	Speed. [km/h]
<i>X</i>	X position. [m]
<i>Y</i>	Y position. [m]
<i>Z</i>	Z position. [m]
<i>yaw</i>	Yaw angle (rotation around the vertical axis of the vehicle). [rad]
<i>pitch</i>	Pitch angle (rotation around the horizontal axis of the vehicle). [rad]
<i>roll</i>	Roll angle (rotation around the horizontal axis of the vehicle). [rad]
<i>vyaw</i>	Yaw angle velocity [rad/s] for vehicle's center of gravity.
<i>vpitch</i>	Pitch angle velocity [rad/s] for vehicle's center of gravity.
<i>vroll</i>	Roll angle velocity [rad/s] for vehicle's center of gravity.
<i>ayaw</i>	Yaw angle acceleration [rad/s ²] for vehicle's center of gravity
<i>apitch</i>	Pitch angle acceleration [rad/s ²] for vehicle's center of gravity.
<i>aroll</i>	Roll angle acceleration [rad/s ²] for vehicle's center of gravity.
<i>ay</i>	Acceleration in vehicle longitudinal direction
<i>ax</i>	Acceleration in vehicle transverse direction
<i>CamS</i>	Altitude of camera above center of gravity (for DPUSGEView)
<i>CamDelta</i>	Pitch angle of camera (for DPUSGEView)
<i>ViewYaw</i>	Yaw angle of camera (for DPUSGEView)

5. Example:

The DPUHeli can be integrated as following:

```
DPUHeli ~Heli
{
    Index = 20;
    Computer = {VDYN};
};

Connections = {
    VDyn.Gear <-> ~Heli.Gear
    ,VDyn.SteeringWheel <-> ~Heli.SteeringWheel
    ,VDyn.AcceleratorPedal <-> ~Heli.AcceleratorPedal
    ,VDyn.BrakePedal <-> ~Heli.BrakePedal

    ,~Heli.X <-> VDyn.X
    ,~Heli.Y <-> VDyn.Y
    ,~Heli.Z <-> VDyn.Z
    ,~Heli.yaw <-> VDyn.yaw
    ,~Heli.pitch <-> VDyn.pitch
    ,~Heli.roll <-> VDyn.roll
    ,~Heli.v <-> VDyn.v
    ,~Heli.v_kmh <-> VDyn.v_kmh
    ,~Heli.vyaw <-> VDyn.vyaw
}
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
,~Heli.vpitch          <-> VDyn.vpitch
,~Heli.vroll           <-> VDyn.vroll
,~Heli.ax              <-> VDyn.ax
,~Heli.ay              <-> VDyn.ay
,~Heli.ayaw            <-> VDyn.ayaw
,~Heli.apitch          <-> VDyn.apitch
,~Heli.aroll           <-> VDyn.aroll

,VDyn.PitchRoad        <-> ~Heli.PitchRoad
,VDyn.RollRoad         <-> ~Heli.RollRoad
,VDyn.XOffset          <-> ~Heli.XOffset
,VDyn.YOffset          <-> ~Heli.YOffset
,VDyn.ZOffset          <-> ~Heli.ZOffset
};
```